STPM Physics · Semester 1 · Chapter 3

Dynamics

3.1 Newton's Laws of Motion

Definitions & laws

- **Newton's first law**: a body remains at rest or moves with constant velocity in a straight line unless acted on by a resultant external force. **Inertia** is the property of a body that resists any change in its state of rest or uniform motion; it is measured by mass.
- **Newton's second law**: the rate of change of momentum of a body is directly proportional to the resultant force acting on it and takes place in the direction of the force: $F = \frac{d(mv)}{dt}$.
- **One newton** is the resultant force that gives a mass of 1 kg an acceleration of 1 m s^{-2} in the direction of the force.
- **Newton's third law**: if body A exerts a force on body B, then B exerts an equal and opposite force on A. The pair acts on different bodies, is of the same type and acts along the same line at the same time.
- **Weight** is the gravitational force on a body, W = mg, directed toward the centre of the Earth; **mass** is the amount of matter and is a scalar that does not change with place.
- **Apparent weight** is the normal reaction exerted on a body, e.g. by a lift floor; it differs from the true weight when the body accelerates vertically.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Newton's second law	$F = ma$	F = ΣF = resultant force (N); m = mass (kg), constant; a = acceleration (m s^{-2})
Constant-mass form of the law	$F = \frac{mv - mu}{t}$	F = force (N); m = mass (kg); u, v = initial and final velocities (m s^{-1}); t = time (s)
Lift accelerating upward	$R = m(g + a)$	R = reaction force on the person (N); m = mass (kg); g = gravitational acceleration (m s^{-2}); a = acceleration of lift (m s^{-2})
Lift accelerating downward	$R = m(g - a)$	R = reaction force on the person (N); m = mass (kg); g = gravitational acceleration (m s^{-2}); a = acceleration of lift (m s^{-2})
Free fall of the lift	$R = 0$	R = reaction force (N); apparent weightlessness
Connected bodies, common acceleration	$a = \frac{(m_1 - m_2)g}{m_1 + m_2}$	a = common acceleration (m s^{-2}); m_1, m_2 = masses (kg); g = gravitational acceleration (m s^{-2}); light string over a smooth pulley

Quantity / law	Formula	Symbols, units & conditions
Tension in connecting string	$T = \frac{2m_1m_2g}{m_1 + m_2}$	T = tension (N); m_1, m_2 = masses (kg); g = gravitational acceleration (m s^{-2}); light string over a smooth pulley
Smooth incline	$a = g \sin\theta$; $N = mg \cos\theta$	a = acceleration down the slope (m s^{-2}); θ = angle of incline; N = normal reaction (N); m = mass (kg); g = gravitational acceleration (m s^{-2})

How to answer (MPM style)

- Draw a free-body diagram showing all forces on one body, label each (W, N, T, f, applied force) and choose positive direction along the acceleration.
- Apply ΣF = ma to each body separately; for connected bodies add equations so internal tensions cancel.
- Write the third-law pair in words: force of A on B and force of B on A, same type, equal magnitude, opposite direction, different bodies.
- Express the result with unit (N) and direction.

Section A traps

Trap: Weight and normal reaction on a block resting on a table are not a third-law pair: they act on the same body and are different types. The pair of weight is the pull of the block on the Earth.

Trap: Forgetting resultant: with a constant velocity ΣF = 0, so a moving body does not need a net force; a net force is needed only for acceleration.

Trap: Lift problems: moving upward at constant speed gives R = mg; acceleration direction, not velocity direction, determines R.

Trap: Inertia gives the effect on a passenger: when a bus brakes the passenger lurches forward, when it starts the passenger moves backward relative to the bus.

Tricks

Trick: Equal-and-opposite forces on one body mean equilibrium (first law), not third law.

Trick: For a connected system, take the total mass for ΣF = (Σm)a, then find tension from one body only.

Trick: Apparent weight greater than mg means a is upward (direction of the net force), whatever the direction of motion.

3.2 Linear Momentum

Definitions & laws

- **Linear momentum** of a body is the product of its mass and velocity, p = mv; a vector with the direction of the velocity, unit kg m s^{-1} or N s.
- **Principle of conservation of linear momentum**: in an isolated system (no resultant external force), the total linear momentum before an interaction equals the total linear momentum after it.
- **Impulse** is the product of the force and the time for which it acts, and equals the change in linear momentum of the body; it is a vector, unit N s.
- Newton's second law (momentum form): the resultant force equals the rate of change of linear momentum.
- A **force–time graph**: the area under the graph equals the impulse (change in momentum); the average force = impulse/time of contact.
- Momentum is conserved in all collisions and explosions of an isolated system, whether elastic or inelastic.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Linear momentum	$p = mv$	p = momentum (kg m s^{-1}); m = mass (kg); v = velocity (m s^{-1})
Conservation of momentum	$\Sigma p(\text{before}) = \Sigma p(\text{after});\ m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$	m_1, m_2 = masses (kg); u_1, u_2 = initial velocities and v_1, v_2 = final velocities (m s^{-1}); one direction taken positive
Impulse (constant force)	$J = F\Delta t = mv - mu = \Delta p$	J = impulse (N s); F = constant force (N); Δt = time (s); u, v = initial and final velocities (m s^{-1}); Δp = change in momentum (kg m s^{-1})
Impulse (varying force)	$J = \int F\,dt$	J = impulse (N s) = area under the F–t graph; F = force (N); t = time (s)
Force as rate of change of momentum	$F = \frac{\Delta p}{\Delta t} = \frac{mv - mu}{\Delta t}$	F = force (N); Δp = change in momentum (kg m s^{-1}); Δt = time interval (s); m = mass (kg); u, v = initial and final velocities (m s^{-1})
Continuous stream hitting a surface	$F = \frac{dm}{dt}(v - u)$	F = force (N); $\frac{dm}{dt}$ = mass flow rate (kg s^{-1}); u, v = velocities before and after (m s^{-1})
Rebound from a wall	$\Delta p = m(v + u)$	Δp = magnitude of change in momentum (kg m s^{-1}); m = mass (kg); u = speed of approach; v = speed of return (m s^{-1})
Explosion or recoil from rest	$m_1v_1 + m_2v_2 = 0;\ v_2 = -\frac{m_1v_1}{m_2}$	m_1, m_2 = masses (kg); v_1, v_2 = velocities after the explosion (m s^{-1})

Quantity / law	Formula	Symbols, units & conditions
Kinetic energy and momentum	$E_k = \frac{p^2}{2m}$	E_k = kinetic energy (J); p = momentum (kg m s^{-1}); m = mass (kg)

How to answer (MPM style)

- Define the positive direction, write the momentum equation with signs, substitute and solve; give the velocity with unit and direction.
- For impulse questions use J = Δp = m(v − u) with the sign of u and v; then $F = \frac{J}{\Delta t}$.
- For "explain" answers: state that the total momentum is conserved because there is no external resultant force on the system.
- Always say that in a force–time graph the area is the impulse and read the value with units.

Section A traps

Trap: Treating momentum as a scalar: an object that rebounds changes momentum by m(v + u), not m(v − u) with magnitudes only.

Trap: Believing momentum is conserved only in elastic collisions; momentum is always conserved in an isolated system, kinetic energy is not.

Trap: Longer contact time gives a smaller average force for the same change of momentum (airbag, crumple zone, bending knees on landing).

Trap: Using the total energy or speed instead of momentum for recoil: for equal momentum magnitudes, the lighter body has the larger speed and the larger kinetic energy.

Tricks

Trick: For recoil from rest, the speeds are in inverse ratio of the masses, $\frac{v_1}{v_2} = \frac{m_2}{m_1}$.

Trick: If two bodies have equal momentum magnitudes, the one of smaller mass has the larger kinetic energy ($E_k = \frac{p^2}{2m}$).

Trick: Area under F–t is impulse; if the graph is a triangle, $J = \frac{1}{2}F_{max} \cdot \Delta t$.

3.3 Elastic and Inelastic Collisions

Definitions & laws

- **Elastic collision** is a collision in which both total linear momentum and total kinetic energy are conserved.

- **Inelastic collision** is a collision in which total linear momentum is conserved but total kinetic energy is not conserved (some is converted to heat, sound and deformation).
- **Perfectly (completely) inelastic collision** is an inelastic collision in which the bodies stick together and move with a common velocity; the maximum loss of kinetic energy consistent with momentum conservation.
- In an elastic collision in one dimension, the relative speed of approach equals the relative speed of separation.
- In a two-dimensional collision, the total momentum is conserved separately along each of two perpendicular axes.
- In an explosion, momentum is conserved and kinetic energy increases.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Conservation of momentum	$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$	m_1, m_2 = masses (kg); u_1, u_2 = velocities before collision (m s^{-1}); v_1, v_2 = velocities after collision (m s^{-1}); holds for both elastic and inelastic collisions
Elastic collision: kinetic energy	$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$	Elastic collisions only; total kinetic energy (J) is conserved
Elastic collision: velocities	$u_1 - u_2 = v_2 - v_1$	Elastic collisions only; u = velocities before, v = velocities after (m s^{-1})
Perfectly inelastic collision	$m_1u_1 + m_2u_2 = (m_1 + m_2)v$	Perfectly inelastic collision; v = common velocity after collision (m s^{-1}); masses in kg
Elastic, m_2 initially at rest	$v_1 = \frac{(m_1 - m_2)u_1}{m_1 + m_2}$; $v_2 = \frac{2m_1u_1}{m_1 + m_2}$	Elastic collision with m_2 initially at rest; v_1, v_2 = final velocities (m s^{-1}); u_1 = initial velocity of m_1 (m s^{-1})
Loss of kinetic energy	Loss = E_k(before) − E_k(after); fractional loss = $\frac{\Delta E_k}{E_k(\text{before})}$	E_k = kinetic energy (J); ΔE_k = loss of kinetic energy (J)
Perfectly inelastic, m_2 at rest	$\frac{\Delta E_k}{E_k} = \frac{m_2}{m_1 + m_2}$	Perfectly inelastic collision with m_2 at rest; $\frac{\Delta E_k}{E_k}$ = fractional loss of kinetic energy
Two-dimensional collisions	Σp_x before = Σp_x after; Σp_y before = Σp_y after	p = momentum (kg m s^{-1}); conservation applied along x and along y separately

How to answer (MPM style)

- Write momentum conservation first with a sign convention, solve for the unknown velocity, then compute kinetic energies before and after to classify the collision.
- Classification wording: "kinetic energy before = kinetic energy after, so the collision is elastic" or "kinetic energy decreases, so inelastic".

- For ballistic pendulum: momentum conservation for the collision, then energy conservation (½(M + m)v^2 = (M + m)gh) for the swing.
- State the lost kinetic energy as heat, sound and deformation when explaining.

Section A traps

Trap: Applying kinetic energy conservation to every collision; only elastic collisions conserve kinetic energy.

Trap: Equating inelastic with sticking together: only a perfectly inelastic collision has a common final velocity.

Trap: Equal masses, head-on elastic collision with one at rest: the moving body stops and the other moves off with the original velocity; this does not hold for an inelastic collision.

Trap: In ballistic pendulum, using energy conservation across the collision, which is inelastic.

Tricks

Trick: Check whether the relative speed of separation equals the relative speed of approach; this decides elastic quickly without calculating energies.

Trick: In an elastic head-on collision with m_1 much smaller than a heavy target, m_1 rebounds with nearly the same speed; with m_1 much larger, it continues almost unchanged.

Trick: Elastic oblique collision of equal masses (target at rest): the final velocities are perpendicular.

3.4 Friction

Definitions & laws

- **Friction** is the force that opposes the relative motion, or the tendency of relative motion, between two surfaces in contact; it acts parallel to the surfaces.
- **Static friction** acts when there is no relative motion; it equals the applied force up to a maximum value (limiting friction) $F_{max} = \mu_s N$.
- **Kinetic (dynamic) friction** acts when the surfaces slide; it is approximately constant and independent of speed, $F = \mu_k N$, and usually less than the limiting static friction ($\mu_k < \mu_s$).
- **Coefficient of friction** is the ratio of the friction force to the normal reaction, $\mu = \frac{F}{N}$; it has no unit and depends on the nature of the surfaces.
- Laws of friction: friction is proportional to the normal reaction; it is independent of the area of contact (for the same normal reaction) and, for kinetic friction, independent of speed.
- **Normal reaction** N is the perpendicular contact force exerted by a surface, not always equal to mg.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Static friction	$F_s \leq \mu_s N$; limiting friction $F_{max} = \mu_s N$	F_s = static friction (N); μ_s = coefficient of static friction; N = normal reaction (N); F_{max} = limiting friction (N)
Kinetic friction	$F_k = \mu_k N$	F_k = kinetic friction (N); μ_k = coefficient of kinetic friction; N = normal reaction (N)
Normal reaction, pulled block	$N = mg - P \sin \theta$	Block on horizontal surface pulled by force P at angle θ above the horizontal; m = mass (kg); g = acceleration of free fall (m s^{-2}); P = pulling force (N)
Horizontal motion, pulled block	$\Sigma F_x = P \cos \theta - \mu N = ma$	Block on horizontal surface pulled by force P at angle θ above the horizontal; μ = coefficient of friction; a = acceleration (m s^{-2})
Normal reaction on incline	$N = mg \cos \theta$	Incline of angle θ; m = mass (kg); g = acceleration of free fall (m s^{-2})
Weight component down slope	$mg \sin \theta$	Incline of angle θ; component of weight down the slope (N)
Point of sliding (limiting angle)	$\tan \theta = \mu_s$	On the point of sliding; θ = angle of friction; μ_s = coefficient of static friction
Sliding at constant speed	$\tan \theta = \mu_k$	Sliding down the incline at constant speed; μ_k = coefficient of kinetic friction
Acceleration down rough incline	$a = g(\sin \theta - \mu_k \cos \theta)$	a = acceleration down the slope (m s^{-2}); g = acceleration of free fall (m s^{-2}); θ = angle of incline; μ_k = coefficient of kinetic friction
Acceleration up slope when pushed	$a = -g(\sin \theta + \mu_k \cos \theta)$	Body pushed up the slope; negative sign indicates deceleration (m s^{-2}); other symbols as above
Stopping distance on rough level surface	$s = \frac{u^2}{2\mu g}$	Body sliding on a rough level surface; s = stopping distance (m); u = initial speed (m s^{-1}); μ = coefficient of friction; g = acceleration of free fall (m s^{-2})
Deceleration on rough level surface	$a = \mu g$	Body sliding on a rough level surface; a = deceleration (m s^{-2})

How to answer (MPM style)

- Draw the free-body diagram with friction opposite to the (impending) relative motion and N perpendicular to the surface.
- Resolve forces parallel and perpendicular to the surface; find N first, then friction from μN, then apply ΣF = ma.

- State whether friction is static or kinetic; if the applied force is below $\mu_s N$, state that the body stays at rest and friction equals the applied force.
- Give answers with unit, and state the direction of friction.

Section A traps

Trap: Using $F = \mu N$ with $N = mg$ when the force has a vertical component or the surface is inclined: N changes ($N = mg \cos \theta$ on an incline).

Trap: Static friction is not always $\mu_s N$; it takes the value that balances the applied force, up to the limit.

Trap: Larger area of contact does not increase friction (same normal reaction); heavier body gives larger friction because N is larger.

Trap: Mixing μ_s and μ_k: the force to start the motion uses μ_s, the force to keep it moving at constant speed uses μ_k.

Tricks

Trick: For an incline, μ is independent of mass: a body slides at the same acceleration for any mass.

Trick: Net force on a sliding body = applied force $- \mu_k N$; if the applied force does not exceed $\mu_s N$, the body stays at rest.

Trick: If $\tan \theta$ is less than μ_s, the block does not slide however large its mass.